HEATMAP NODE · REGULATORY REGISTER

HeatMap Node regulatory and standards register

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DOCUMENT CONTROL

Document ID	HMN-REG-001	Version	0.1
Status	Draft	Date	2026-11-01
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/07-regulatory.md @ b9c67bc		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First register at the kit 1.9.1 upgrade; sources checked 2026-11-01; anything marked to assess needs a qualified reviewer

HeatMap Node regulatory and standards register

Which standards and regulations bear on a street-pole heat stress node that carries a solar cell, a LoRa radio and no personal data. Not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Nothing here is certified: the node is a research and educational prototype at TRL 3. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	ISO 7243:2017, WBGT index for heat stress	International	Definition of the wet bulb globe temperature index and its weights (0.7 natural wet bulb, 0.2 globe, 0.1 air outdoors with solar load)	Defines the quantity the node reports (PR-005); the node supplies the index as data and does not assess exposure	Design lead	Applies; met by design (HMN-CAL-001 B2)
G2	ISO 7726, instruments for measuring physical quantities of the thermal environment	International	Accuracy classes and response times of globe, air and humidity instruments	Reference for the 150 mm globe, the air temperature class and response; edition and class limits to check against the paper	Design lead	To assess
G3	NIOSH Publication 2016-106, criteria for occupational exposure to heat	United States	Recommended exposure limits that use WBGT with work rate and acclimatization	Context for users; the node is not an exposure-limit instrument	Design lead	Does not apply to the product; context only
G4	OSHA heat exposure guidance	United States	Employer programs for heat	Employers may read the data; the node is not a compliance instrument	Partner	Does not apply; to assess before any claim to employers
G5	47 CFR 15.247 (FCC), digital modulation and hopping systems in 902 to 928 MHz	United States	Dwell time and power for the LoRa radio	One uplink of 247 ms at SF9 is under the 400 ms dwell (STEM S16); the radio module's grant is FieldNode's	FieldNode lead	Applies; met by design at the HeatMap level; module certification to assess in FieldNode
G6	LoRa Alliance RP002-1.0.4 regional parameters	US915 plan	Channels, data rates, dwell	Fixes the airtime budget (HMN-CAL-001 G2)	Design lead	Applies; met by design
G7	47 CFR Part 15 Subpart B, unintentional radiators	United States	Emissions from the electronics (fan, MCU)	The fan and sensor board are inside FieldNode's emissions envelope	FieldNode lead	To assess
G8	IEC 60529, degrees of protection (IP code)	International	IP65 for the enclosure	FieldNode's enclosure carries IP65 (PR-008); HeatMap adds M12 sealed leads and a	Design lead	Met by design; certification needed later

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
				PTFE vent cap on the SHT45		
G9	UN Manual of Tests and Criteria, Part III, 38.3 (lithium cells)	International transport	Cell safety tests for shipping	Applies to cells shipped inside FieldNode; the cell supplier's report is needed	FieldNode lead	Certification needed later
G10	RoHS 2011/65/EU, WEEE 2012/19/EU, Radio Equipment Directive 2014/53/EU	European Union	Substances, end-of-life, radio conformity if the node is sold in the EU	Not for a prototype; relevant only if a partner sells in the EU	Partner	To assess
G11	AASHTO structural supports for highway signs, luminaires and traffic signals	United States	Wind loads and fatigue of attachments to light and signal poles	Pole owners may ask for the node's wind area and moment; the head's 35 m/s case is in HMN-CAL-001 H1 to H6	Pole owner	To assess
G12	Pole-owner attachment permission (city or utility)	Local	Permission and liability for clamped attachments	Needed for every pole; no drilling	Partner	Applies; to assess per city
G13	Personal data law (for example GDPR in the EU, state privacy laws in the US)	Local	Personal data	The node holds environmental data only (PR-012); no camera, microphone or radio scanning, so personal data law is expected not to apply	Design lead	Does not apply by design; to assess by a lawyer before deployment
G14	WMO Guide to Instruments and Methods of Observation (WMO-No. 8)	International guidance	Radiation shields, siting and exposure of air temperature sensors	Street siting breaks the open-site rule on purpose; the node is a street-level instrument, not a climate station	Design lead	Guidance; does not apply to the product

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

All accessed 2026-11-01.

- ISO 7243:2017, https://www.iso.org/standard/67188.html.
- NIOSH 2016-106, https://www.cdc.gov/niosh/docs/2016-106/default.html.

- OSHA, "Heat exposure," https://www.osha.gov/heat-exposure.
- FCC, 47 CFR 15.247, https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15/subpart-C/subject-group-ECFR2a5a6f3d3f3e4a6/section-15.247; 47 CFR Part 15, https://www.ecfr.gov/current/title-47/part-15.
- LoRa Alliance, RP002-1.0.4 Regional Parameters, https://resources.lora-alliance.org/technical-specifications/rp002-1-0-4-regional-parameters.
- IEC, IP ratings, https://www.iec.ch/ip-ratings.
- UNECE, UN Manual of Tests and Criteria, https://unece.org/transport/dangerous-goods/un-manual-tests-and-criteria.
- AASHTO store, structural supports specification, https://store.transportation.org/Item/CollectionDetail?ID=205.
- WMO, CIMO Guide, https://www.wmo.int/pages/prog/www/IMOP/CIMO-Guide.html.
- ISO 7726, EU directives (RoHS, WEEE, RED) and the privacy laws: listed by name only; not retrieved in this session, so editions and obligations are "to assess".